Q). Find out the list of companies which will ask DSA for placement drive.

Product-based companies:

1. Google
2. Microsoft
3. Amazon
4. Adobe
5. Atlassian
6. Flipkart
7. Walmart Global Tech
8. Razorpay
9. Oracle
10. SAP
11. Service Now.

Service-based companies:

1. TCS
2. Infosys
3. Accenture
4. WIPRO
5. Cognizant
6. Capgemini
7. HCL Tech
8. Tech Mahindra.

Q. Find out the gate Syllabus for DSA.

Unit 1: Programming and Data Structures

1. Introduction to Data Structures
2. Arrays
3. Linked Lists
4. Stacks
5. Queues
6. Hash Tables (Hashing)
7. Trees
8. Graphs
9. Recursion

Unit 2: Algorithms

1. Asymptotic Analysis
2. Searching Algorithms
3. Sorting Algorithms
4. Hashing
5. Divide and conquer
6. Greedy Algorithms
7. Dynamic programming
8. Graph Algorithm.

Q). Find out set of input and output for the following :-

1) Electricity Bill

Input
- Consumer Number
- Consumer Name
- Service number
- Meter number
- Previous Meter reading
- Tariff type
- Rate per unit
- Fixed charges

output
- Consumer details
- units consumed
- Energy charges
- Fixed charges
- Electricity Duty/Tax
- Total Bill amount
- Due Date
- Bill status.

2) Bus ticket :

Input
- Passenger Name
- Passenger Age
- Grender
- Source
- Destination
- Tourney Date
- Bus Type

output
- Ticket Number
- passenger details
- Bus number
- Tourney details
- Seat number
- Departure time
- Arrival time.

3). Message to a friend

Input	output
Sender Name	Delivered message
Receiver Name	Menage status
Contact Number	Date & Time Sent
Message text	Receiver details
Date	Confirmation
Time	notification.

Q) Today when you started from home to college, what are the various data structures identified in your surroundings.

	Data Structure	Real-life Ex
1.	Array	Houses in a street
2.	Linked List	people standing in a queue
3.	stack	stack of plates
4.	Queue	ticket counter Queue
5.	Tree	college organis Chart
6.	Set	Unique studen in a class.